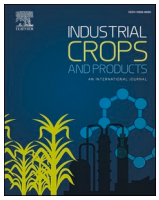
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Rapid microwave-assisted pretreatment using a ternary deep eutectic solvent for lignin fractionation of eucalyptus

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ABSTRACT

This work developed a strategy for rapid microwave-assisted refining of eucalyptus wood in 30 min using a ternary DES (lactic acid: choline chloride: polyethylene glycol 400, 10:1 + 5%) with an excellent delignification rate of 89.73%. The test of Kamlet−Taft solvatochromic parameters of DESs shows that the acidity and polarity of the solution greatly influence the delignification rate. The polyethylene glycol 400 increased the solubility of lignin in the solvent, improved the delignification rate, and protected cellulose from degradation. The gel permeation chromatography, 2D-HSQC NMR, and ^{31}P NMR analysis show that the lignin obtained after DES treatment had low molecular weight, small polydispersity index, and relatively rich content of β-O-4 and phenolic hydroxyl groups, which has great application potential in high-value-added products from lignin. At the same time, the main products of the simulated reaction of the lignin model compound were monophenols and Hibbert ketone, and the possible reaction mechanism of changes in lignin during treatment was given. Our study lays a foundation for applying DES in biomass refining and the high-value utilization of lignin.

1. Introduction

The depletion of fossil fuels and pressing environmental concerns have engendered a significant inclination toward the pursuit of sustainable and renewable resources. Lignocellulosic biomass, characterized by its abundance and renewability, holds significant potential as a viable substitute for fossil-based feedstock. This biomass can be effectively transformed into both fuels and chemicals, offering a promising avenue for sustainable energy production and resource utilization (Liu et al., 2019; Rana et al., 2022, 2023; Rana, 2022). The primary constituents of lignocellulose are cellulose, hemicellulose, and lignin. Lignin is the wealthiest resource of aromatic compounds in nature, and there are various applications for the utilization of this substance in the production of high-value fine chemicals, fuel products, and functional materials (Shen et al., 2019a). However, removing lignin from lignocellulosic biomass is difficult due to its complicated internal network structure and stubborn lignin.

To overcome the inherent stubbornness of lignocellulose, researchers have developed different pretreatment methods to upgrade lignocellulose biomass into high-value products (Haghighi Mood et al., 2013). However, the strong corrosive nature of diluted acid, the high energy and high-pressure requirements of steam explosion and hydrothermal pretreatment, the high volatility of organic solvents used in organic solvent pretreatment, and the high cost, toxicity, and poor biodegradability of ionic liquids are some drawbacks that are typically associated with these techniques (Ci et al., 2020).

Deep eutectic solvents (DESs), eutectic mixtures with a lower eutectic temperature than the ideal liquid combination, have recently received much interest in biomass pretreatment due to their remarkable selectivity for lignin removal (Tan et al., 2018; Alvarez-Vasco et al., 2016; da Costa Lopes et al., 2020). The DES system comprises a minimum of two hydrogen bond donors (HBDs) and hydrogen bond acceptors (HBAs) (Smith et al., 2014). Due to its cheap cost, non-toxicity, biodegradability, and economical production, the quaternary ammonium salt is the most extensively used HBA for DES, specifically choline chloride (ChCl) (Schutyser et al., 2018). ChCl-based DESs have been extensively utilized in diverse fields as an eco-friendly solvent alternative (Smith et al., 2014; Maugeri and de Maria, 2012; Zafarani-Moattar et al., 2016). However, the study of DES pretreatment usually focuses on non-wood (more than 80%) (Zhou et al., 2022; Chen et al., 2020; Tan et al., 2020; Wang and Lee, 2021). A more comprehensive investigation could be conducted regarding the availability of raw wood resources,

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especially focusing on eucalyptus, which is widely cultivated as hardwood and exhibits a significant rate of production (Shen et al., 2019a). Compared with non-wood biomass materials, the dense structure of hardwood is more difficult to deconstruct. Therefore, it usually takes 6–12 h or more to get relatively satisfactory pretreatment results in the few studies on DES pretreatment of eucalyptus (Hong et al., 2022; Shen et al., 2019b; Chen et al., 2021; Soares et al., 2021). Long pretreatment time and the acidity of DES will lead to lignin condensation and reduce the efficiency of enzyme treatment (Hong et al., 2020a). In addition, long-term pretreatment will lead to various side effects and degradation of DES and cellulose (Morais et al., 2021). Hence, it is imperative to expeditiously execute the delignification process to segregate biomass into superior-grade cellulose fibers and lignin. Microwave pretreatment has been widely employed in many bio-refining processes due to its non-contact heat source, low residence time, and minimal adverse effects (Zhong et al., 2022; Muley et al., 2019). Furthermore, microwave radiation has been demonstrated to improve the ionic properties of DES and raise the molecular polarity (Chen and Wan, 2018). Microwave-assisted deep eutectic solvents (DES) have been proven effective for delignification of non-wood biomass feedstocks. For instance, Yan et al. (2023) treated rice straw continuously for 4 min under microwave irradiation (680 W) at a 1:30 solid to liquid ratio using a ternary DES (choline chloride: oxalic acid: lactic acid 1:0.5:1, molar ratio), yielding a 63.4% lignin removal rate. Chourasia et al (Chourasia et al., 2022). employed a choline chloride/ethylene glycol/$NiCl_2{\cdot}6H_2O$ system (molar ratio 1:2:0.016 with 20 wt% water as co-solvent) to treat sugarcane bagasse at 100°C for 30 min at 10% solid loading, reaching 84% delignification. Ji et al (Ji et al., 2020). utilized a ternary DES (ChCl-Gly-Al at molar ratio 1:2:0.2) under consecutive ultrasound (20 + 28 + 40 kHz, room temperature, 30 min) and microwave (80°C, 20 min) treatment of garlic skin to attain 90.14% lignin removal. Hence, it is plausible that the utilization of microwaves can reduce the duration of the reaction process in DES pretreatment of eucalyptus compared with DES pretreatment alone and achieve similar or even higher efficiency.

This study has devised an eco-friendly, gentle, and effective technique for pretreatment by utilizing a ternary deep eutectic solvent to fractionate lignin from eucalyptus. Furthermore, the potential delignification mechanism was elucidated through an investigation of the structural modifications of lignin pre- and post-treatment, as well as the simulation of the reaction process utilizing lignin model compounds.

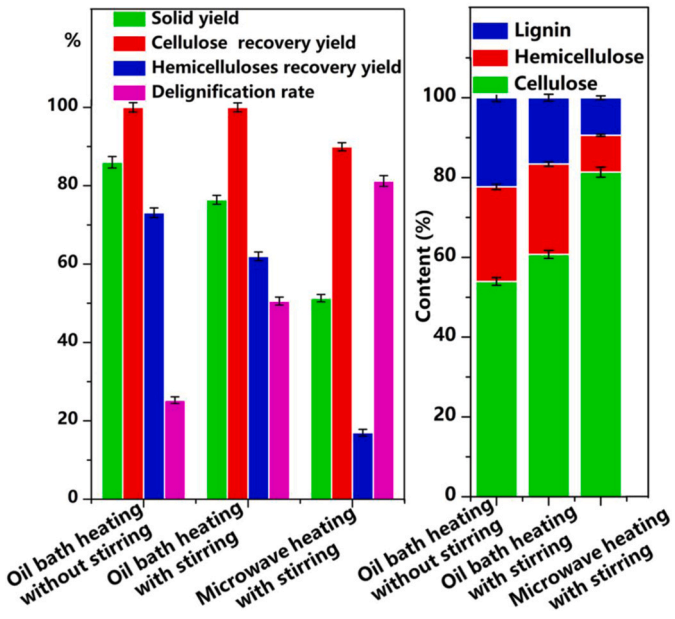


Fig. 1. Pretreatment results and content of SR components using different heating methods. Other conditions: LA:ChCl 10:1, 140℃ 1 h.

2. Experimental materials and methods

2.1. Materials

Samples of Eucalyptus were obtained from a commercial forest farm in Paiyang Mountain, China. The composition of the Eucalyptus was determined to be 25.68% lignin, 46.40% cellulose, and 27.92% hemicellulose, based on the dried extracted raw materials. The specific experimental procedures and relevant formulas could be found in the supply materials (Note. S2). Eucalyptus wood powder passing through 40 mesh sieves was selected as the raw material (RM). The milled wood lignin (MWL) and alkali lignin (AL) were obtained based on the reported routes (Wen et al., 2013; Liu et al., 2023). The support information could include specific steps (Note. S1). The reagents used in this work are analytically pure and purchased from Beijing Innochem Technology Co., Ltd.

2.2. Methods

2.2.1. Preparation of DESs

Several commonly used DESs for biomass pretreatments were prepared. Specific components and proportions can be found in Tab. S1. Specifically, a beaker was filled with a specified molar ratio of HBAs and HBDs. The obtained DESs were then kept at room temperature after being agitated in a water bath at 60°C to produce a clear solvent.

2.2.2. DES pretreatment and DES recycling

The pretreatment process was conducted utilizing a conical flask, wherein the biomass concentration was maintained at 10 wt% (5.00 g of dry biomass and 50.00 g of reagents). The pretreatment was carried out under different target conditions (solvent type, heating mode, time, etc.) The resulting liquid and solid were transferred to a 500 mL beaker containing 250 mL of washing solvent (acetone: ethanol 1:1 v:v) and agitated for 6 h after pretreatment. Following that, the combined solution was filtered through a G2 glass filter, and the solid residue (SR) was washed with washing solvent until the filtrate was colorless. Rotary evaporation was used to concentrate the filtrate, which was subsequently placed into water to precipitate lignin. The SR and lignin obtained from the DES pretreatment (DESL) were dried in a vacuum oven at 40°C. By evaporating the liquid, DES was recovered and reused.

2.2.3. Component analysis of SR

The lignin content in SR was determined by the two-step acid hydrolysis (Shen et al., 2019b), the cellulose content in SR was determined by the nitric acid-alcohol method, and hemicellulose content was calculated by subtraction. The specific experimental procedures and relevant formulas could be found in the supply materials (Note. S2).

2.2.4. Measurement of the Kamlet−Taft (K-T) solvatochromic parameters of DESs

The K-T solvatochromic parameters of DESs used in this work were measured based on reported routes (Dwamena and Raynie, 2020), where α stands for hydrogen bond acidity, the β stands for hydrogen bond alkalinity, and the π^* stands for the polarity of the solvent. The specific steps and relevant formulas can be found in the support information (Note. S3).

2.2.5. Characterizations of lignin

The application of gel permeation chromatography (GPC) was employed to assess the acetylation of materials for the purpose of determining the distribution of molecular weight (Mw) in lignin. The chemical structures of lignin were studied using Fourier transform infrared (FTIR) spectroscopy and 2D-HSQC NMR. The ^{31}P NMR spectra were used to measure the phenolic hydroxyl, alcohol hydroxyl, and carboxyl concentrations of lignin. The specific steps and relevant formulas can be found in the support information (Note. S4).

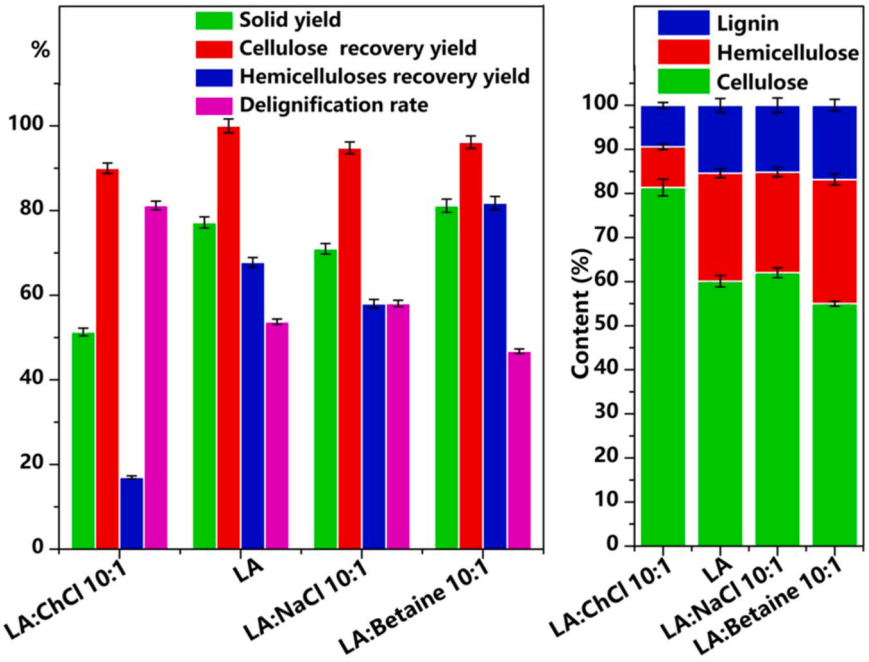


Fig. 2. Pretreatment results and content of SR components using different DES solvent systems. Other conditions: microwave heating at 140℃ for 1 h, the dosage of PEG 400 was 5 g.

2.2.6. Analysis of simulated reactions by model compound

The simulation reaction of lignin in the solvent system was carried out with the typical model compound Guaiacylglycerol-beta-guaiacyl Ether (GBG) under optimal reaction conditions. The reaction product underwent three successive extractions using ethyl acetate and water. Then the extracts were combined and concentrated and sent to GC-MS for qualitative and relative quantitative analysis of the products. GC-MS method for analysis of hydrogenolysis products: the temperature program started at 40℃ for 10 min, heated at 10 ℃/min to 90℃ for 5 min, then heated at 15 ℃/min to 260℃ for 5 min and heated at 15 ℃/min to 290℃ for 0 min. The method of GC-MS acid hydrolysis product analysis: the temperature program started at 45℃ for 0 min, and the temperature was heated at 12℃ to 300 ℃/min and kept for 10 min.

3. Results and discussion

3.1. Analysis of DES pretreatment and content of SR components

3.1.1. The effect of heating mode

The LA-ChCl (LA:ChCl 10:1, molar ratio) was selected as the binary DES in this work based on our previous works (Liu et al., 2020; Cui et al., 2022).

As shown in Fig. 1, the delignification rate was greatly improved by stirring and microwave heating in the solvent system of LA-ChCl. The solvent can contact the raw materials more evenly and frequently by stirring, thus improving the treatment effect. Microwave improved the heating efficiency of the solvent, thus improving the treatment effect. Therefore, the combination of microwave and stirring was a relatively suitable pretreatment method. Fig. S1 shows the digital photos of SR obtained by different heating methods.

3.1.2. The effects of functional groups and K-T parameters

To better understand the impact of functional groups in the LA-ChCl and K-T parameters of the solution on the microwave treatment, the corresponding comparative experiments were carried out.

Fig. 2 shows that HBA greatly influences the DES pretreatment process. Compared with the pretreatment results of LA, the delignification rate in the LA:ChCl 10:1 was greatly improved, which indicated that a small ratio of ChCl played an essential role in the removal of lignin, and this result was similar to the previous report (Hong et al., 2020b). In this work, the function of the choline ion was remarkable. The delignification rate sustainably decreased when ChCl was replaced by NaCl. When ChCl was replaced by betaine, the delignification rate was the lowest, only 46.77%.

Table 1
The K-T parameters and delignification of DESs.

DES	α	β	π*	Delignification rate (%)
LA: ChCl 2:1	1.92	2.59	0.49	43.29
LA:ChCl 10:1	2.25	1.81	1.28	81.20
FA:ChCl 2:1	2.01	1.19	1.06	69.43
FA:ChCl 10:1	2.94	1.59	0.02	58.25
AA:ChCl 10:1	1.87	1.76	0.95	13.21
PA:ChCl 10:1	1.64	1.93	0.93	12.34

The other conditions: microwave heating at 140℃ for 1 h.

The previous report (Kanbayashi and Miyafuji, 2016) shows that ionic liquid based on the choline group can expand and rupture the cell wall, and the expansion of wood will increase its permeability. Consequently, the diffusion rate of LA into the cell wall will be accelerated, while the diffusion rate of lignin out of the cell wall will be enhanced, thus improving delignification efficiency. Our previous research (Liu et al., 2023, 2020) shows that choline ion directly participated in the reaction between solvent and lignin because N element and new hydrogen spectrum signals were detected in lignin obtained after treatment. Fig. S2 shows the digital photos of SR processed by DES with different HBA.

According to Table 1 and transformation based on LA structure (Fig. S6), the hydroxyl group in LA increased the polarity and proton-donating ability of the solution, thus increasing the delignification rate (LA:ChCl 10:1 > PA:ChCl 10:1). Carboxyl groups improved cleavage of lignin bonds because ethanol:ChCl 10:1 and isopropanol:ChCl 10:1 were not able to remove lignin at 140℃ for one hour. The increase in alkyl chain length reduced the acidity and delignification rate of DESs (FA:ChCl 10:1 > AA:ChCl 10:1 > PA:ChCl 10:1).

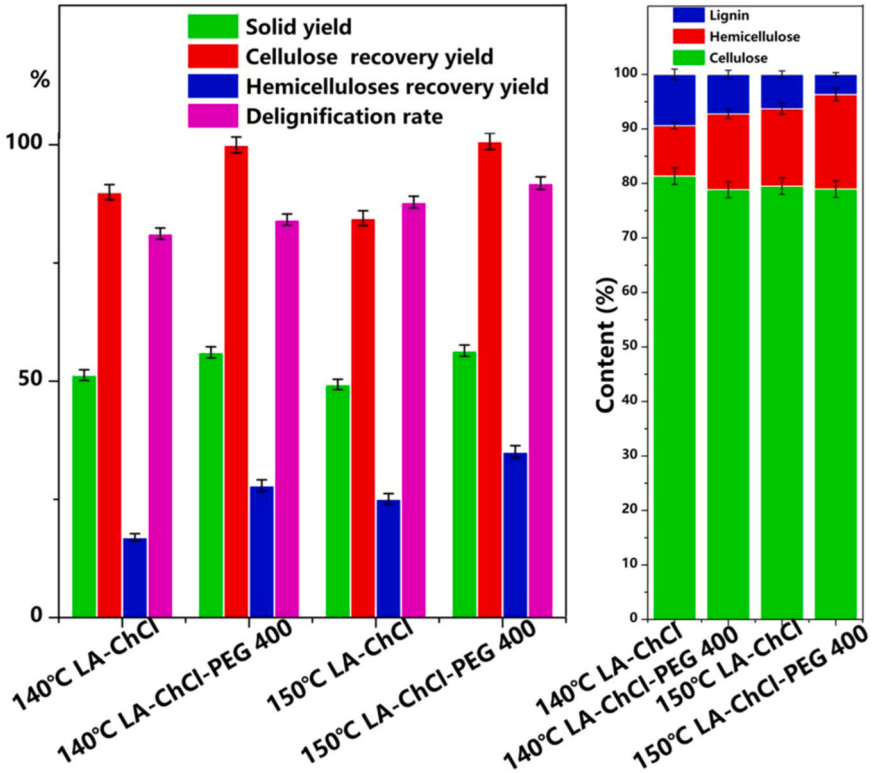


Fig. 3. Pretreatment results (left) and content of SR components (right) using binary and ternary DES. Other conditions: microwave heating for 1 h, the dosage of PEG 400 was 5 g.

In Table 1, the highest delignification rate was LA:ChCl 10:1 (81.20%), followed by FA:ChCl 2:1 (69.43%). The K-T parameters of these two DES were similar, and they both had high polarity and appropriate acidity and alkalinity. The above results show that the polarity and acidity of solvents greatly influenced the microwave-assisted DES pretreatment process. The greater the polarity of the solution, the stronger the effect of microwave on polar molecules; the stronger the motion of polar molecules, the higher the heating efficiency. In addition, according to the similarity compatibility principle, the higher the polarity of the solvent, the easier it was for lignin to dissolve due to the polarity of lignin. Therefore, solvents with high polarity were suitable for the microwave treatment of biomass materials. The acidity of the solution can improve the delignification rate, but solvents with high acidity will seriously degrade cellulose and hemicellulose, thus reducing the selectivity of delignification.

3.1.3. *The effect of PEG 400*

Although the delignification rate in LA-ChCl was excellent, cellulose

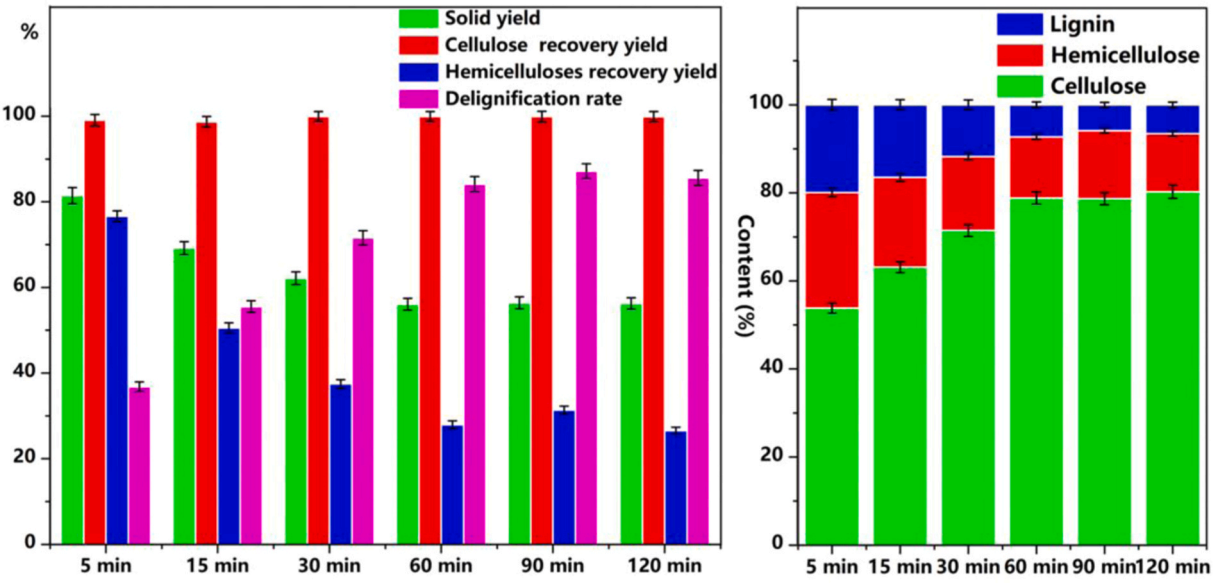


Fig. 4. Ternary DES pretreatment results and content of SR components for different times. Other conditions: LA-ChCl-PEG 400, the dosage of PEG 400 was 5 g, microwave heating at 140℃.

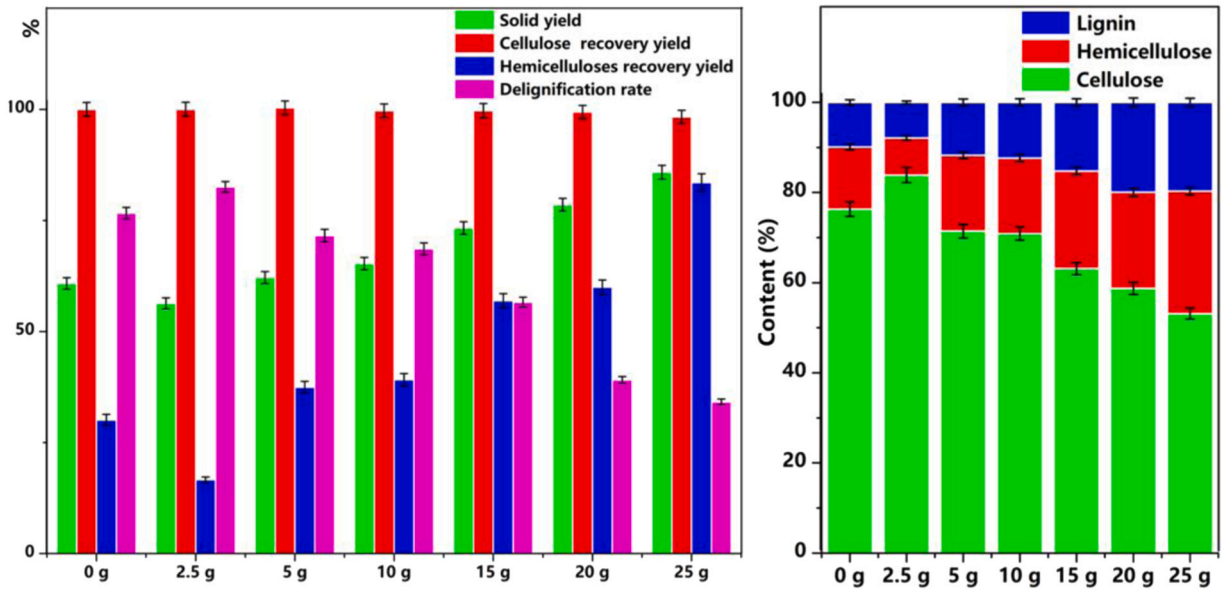


Fig. 5. Ternary DES pretreatment results and content of SR components using different dosages of PEG 400. Other conditions: LA-ChCl-PEG 400, microwave heating at 140℃ for 30 min.

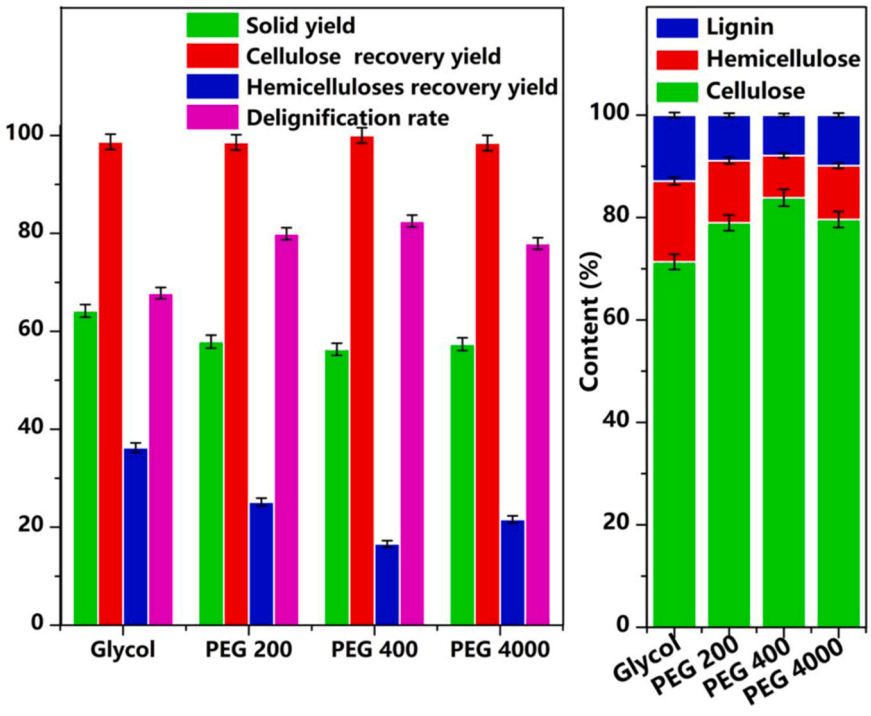


Fig. 6. Ternary DES pretreatment results and content of SR components using different third compositions of DES. Other conditions: LA-ChCl, the dosage of the third component of DES was 2.5 g, microwave heating at 140℃ for 30 min.

was partly degraded due to its acidity (the cellulose recovery yield was 89.99% when heated at 140℃ for 1 h), which was unexpected in the pretreatment process. Polyethylene glycol 400 (PEG 400), the third component, was introduced concerning the method of nitric acid ethanol, in which ethanol can protect cellulose from being degraded by nitric acid. Polyethylene glycol (PEG) is a good solubilizer and phase transfer catalyst. It has long hydrophobic alkyl chains and hydrophilic hydroxyl groups, and its properties are similar to lignin. It has been reported that long-chain diols can increase the dissolution of lignin and inhibit the condensation of lignin (Yu et al., 2022). In addition, PEG is a safe, non-toxic, and very stable solvent (usually considered a secure compound) on the GRAS list of the FDA.

It can be seen from Fig. 3 that the solid yield, the cellulose and hemicellulose recovery yield, and the delignification rate were all improved after adding PEG 400 at 140℃ and 150℃ for 1 h, and the cellulose recovery yield was almost 100%. This shows that the third component increased the dissolution of lignin while protecting cellulose and hemicellulose from degradation and further increased the delignification selectivity of the DES. Fig. S3 shows the digital photos of SR obtained after lactic acid, binary and ternary DES treatment.

After the addition of the third component, the influences of time (Fig. 4), PEG 400 dosage (Fig. 5), the type of the third component

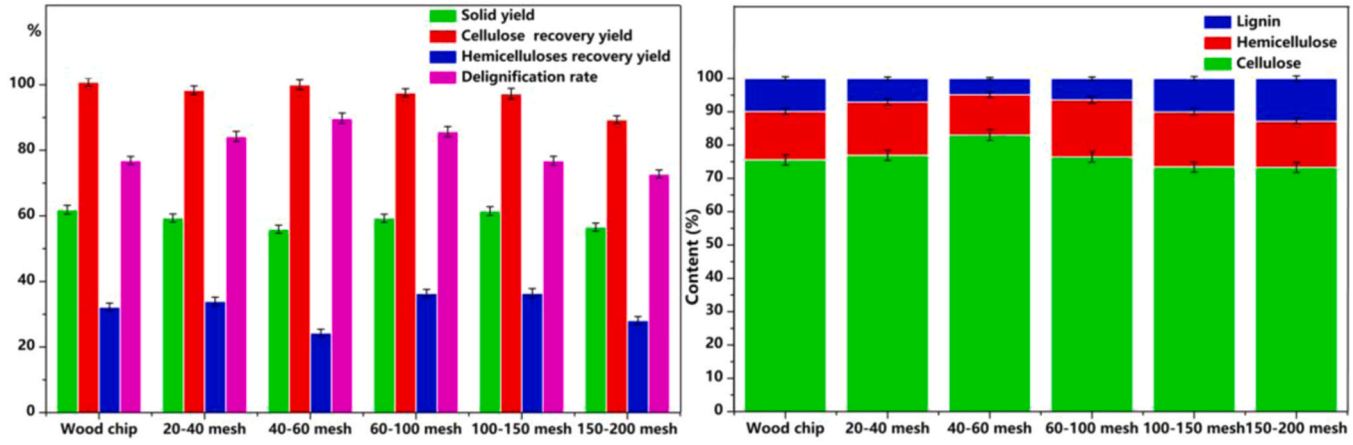


Fig. 7. Ternary DES pretreatment results and content of SR components using different particle sizes of materials. Other conditions: LA-ChCl-PEG 400, the dosage of the third component of DES was 2.5 g, microwave heating at 140℃ for 30 min.

Table 2
Comparative investigations of the delignification rate of wood pretreatment in literature.

DES (HBD:HBA)	Biomass	Conditions: Temperature (℃), and Solid loading (wt%)	Time	Delignification rate (%)	References
LA:ChCl+PEG 400 (10:1 +5%)	Eucalyptus	140℃, 10%	30 min	89.72	This work
LA:ChCl 2:1	Douglas fir	145℃, 10%	6 h	58.2	Alvarez-Vasco et al., 2016
LA:ChCl 2:1	Poplar	145℃, 10%	6 h	78.5	Alvarez-Vasco et al., 2016
LA:ChCl 10:1	Eucalyptus camaldulensis	130℃, 10%	6 h	93.2	Shen et al., 2019a
OA:ChCl:1,5-pentanediol 1:1:1	Poplar	110℃, 6.68%	5 h	89.6	Yu et al., 2022
PA:U+PTSA (1:2 +25%)	Eucalyptus globulus	190℃, 5%	8 h	80.64	Soares et al., 2021
LA:ChCl 10:1	Cunninghamia lanceolata	120℃, 3.3%	12 h	81.73	Wu et al., 2021
LA:ChCl 10:1	Willow	120℃	12 h	91.8	Li et al., 2017

(Fig. 6), and raw material size (Fig. 7) on the pretreatment process were considered.

3.1.4. The effect of time

As shown in Fig. 4, the influence of reaction time highlights the high efficiency of ternary DES pretreatment. A great deal of delignification (36.81%) began in 5 min and reached 55.53% and 71.59% in 15 min and 30 min, respectively. However, the delignification rate increased slightly after 30 min and almost remained unchanged with time increased after one hour. Fig. S4 shows the digital photos of SR obtained by ternary DES processing with different times.

Table 3
GPC analysis of lignin.

Sample	Mw	Mn	PDI
MWL	6359	2306	2.78
LA[a]	7827	3975	2.41
LA-ChCl[a]	4589	2266	2.01
LA-ChCl-PEG 400[a]	5519	2629	2.12
15 min[b]	6097	3035	2.25
30 min[b]	5724	2735	2.20
60 min[b]	5519	2629	2.12
0 g[c]	5726	2764	2.17
2.5 g[c]	5613	2730	2.14
5 g[c]	5724	2735	2.20
Glycol[d]	6096	2984	2.19
PEG 200[d]	5682	2683	2.17
PEG 400[d]	5613	2730	2.14
PEG 4000[d]	6365	3142	2.17
MWL	6359	2306	2.78
DESL*	5613	2730	2.14
AL	7149	2151	3.32

a: microwave, 140℃, 1 h, 5 g PEG 400. b: LA-ChCl-PEG 400, microwave, 140℃, 5 g PEG 400. c: microwave, LA-ChCl-PEG 400, 140℃, 30 min. d: microwave, LA-ChCl-, 140℃, 30 min, 2.5 g. *The DESL was obtained under the following conditions: LA-ChCl-PEG 400, microwave heating at 140℃ for 30 min, and the dosage of PEG 400 was 2.5 g. The same labels were applied in the FTIR analysis.

3.1.5. The effect of dosage of PEG 400

With the increase in the dosage of the third component, the delignification rate first increased and then decreased (Fig. 5), reaching the highest value (82.65%) at 2.5 g. The possible reason is that although the delignification rate can be improved by adding a certain amount of PEG 400, too much PEG 400 will continuously reduce the polarity and acidity of DES (Fig. S7). In addition, the viscosity of the DES increased after adding PEG 400 (Tab. S2). Fig. S5 shows the digital photos of SR obtained by ternary DES treatment with different PEG 400 content.

3.1.6. The effect of chain length of the third component

As shown in Fig. 6. Except for EG, the delignification rate was similar when PEG 200, PEG 400, and PEG 4000 were added, but it was slightly higher when PEG 400 was added. The possible reason is that the properties of long-chain PEG and lignin are similar, and it is easier for them to play a role in the reaction system.

3.1.7. The effect of particle sizes of raw materials

In the DES pretreatment process, the particle sizes of raw materials used in previous reports are different (Hong et al., 2020a; Su et al., 2021; Liu et al., 2017). The size of biomass particles will affect the conduction efficiency between the solution medium and biomass, and it should be considered. Fig. 7 shows the influence of raw material size on the pretreatment process.

The delignification rate first increased and then decreased with the decrease in raw material size and reached the maximum at 40–60 mesh

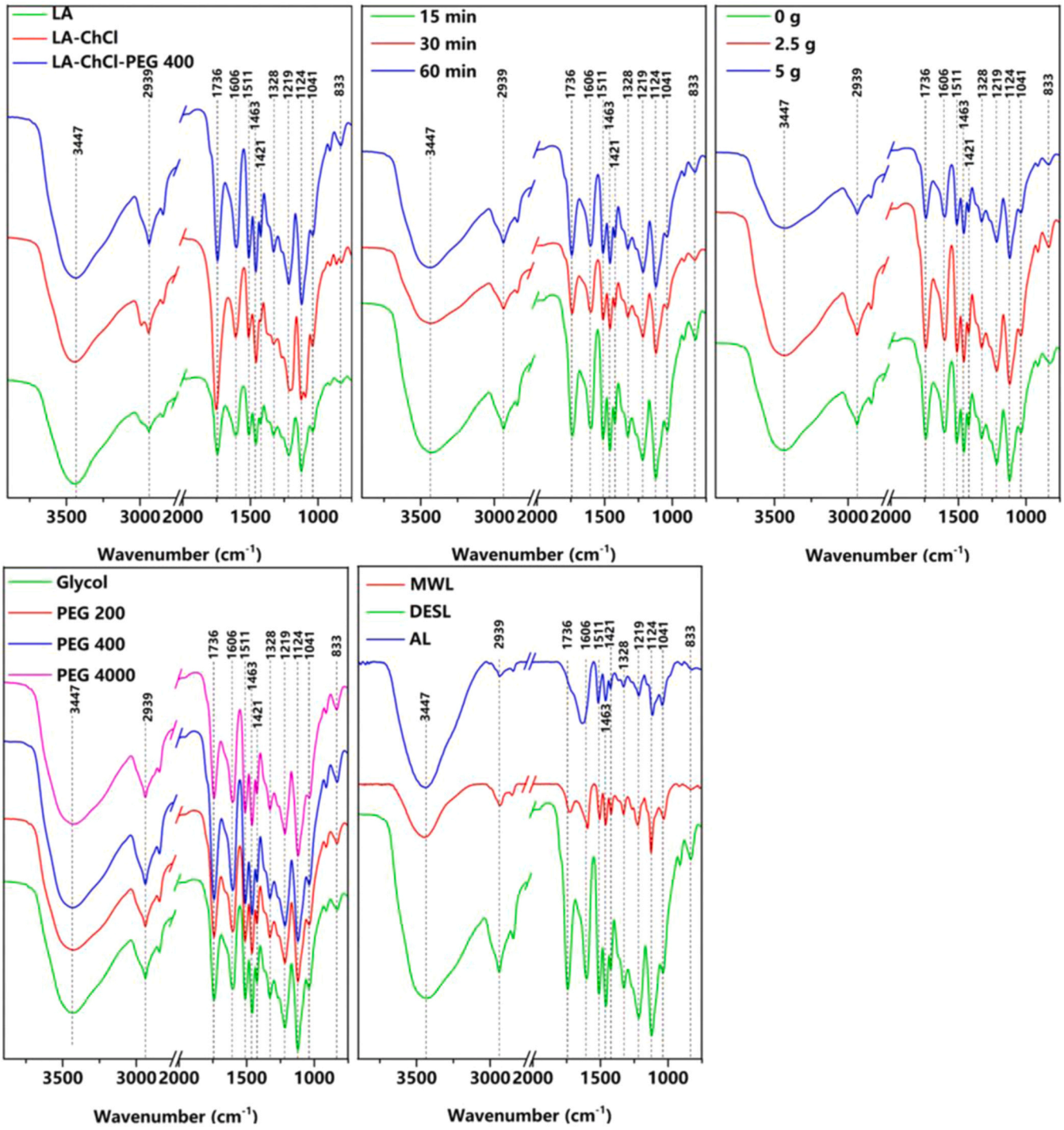


Fig. 8. FTIR spectra of lignin obtained in this work. The DESL was obtained under the following conditions: LA-ChCl-PEG 400, microwave heating at 140℃ for 30 min, and the dosage of PEG 400 was 2.5 g.

(89.72%). The large size of a raw material makes it relatively difficult for the solution to penetrate. In contrast, the small size makes it difficult for the solution to contact the raw material because the gap between raw material particles is too narrow. Compared with other previously reported wood pretreatment work (Table 2), microwave-assisted pretreatment with a ternary DES in this work is very efficient, and 90% lignin can be removed in 30 min at 140℃. Especially when dealing with wood chips, 76.91% of lignin can be removed. Compared with the traditional alkaline pulping (usually cooking at 160–170℃ for 3–4 h, which requires high pressure), the reaction system adopted in this work can cook eucalyptus chips at 140℃ for 30 min under normal pressure, with a pulping yield of 61.89%, showing great potential in pulping.

3.2. Characterizations of lignin

3.2.1. GPC analysis

The average molecular weight (Mw and Mn) and polydispersity index (PDI) of lignin samples were determined by GPC, and the results are presented in Table 3.

In comparison to MWL, the Mw of lignin exhibited an increase subsequent to treatment with LA, potentially attributable to the condensation of lignin within an acidic milieu. The Mw of lignin decreased

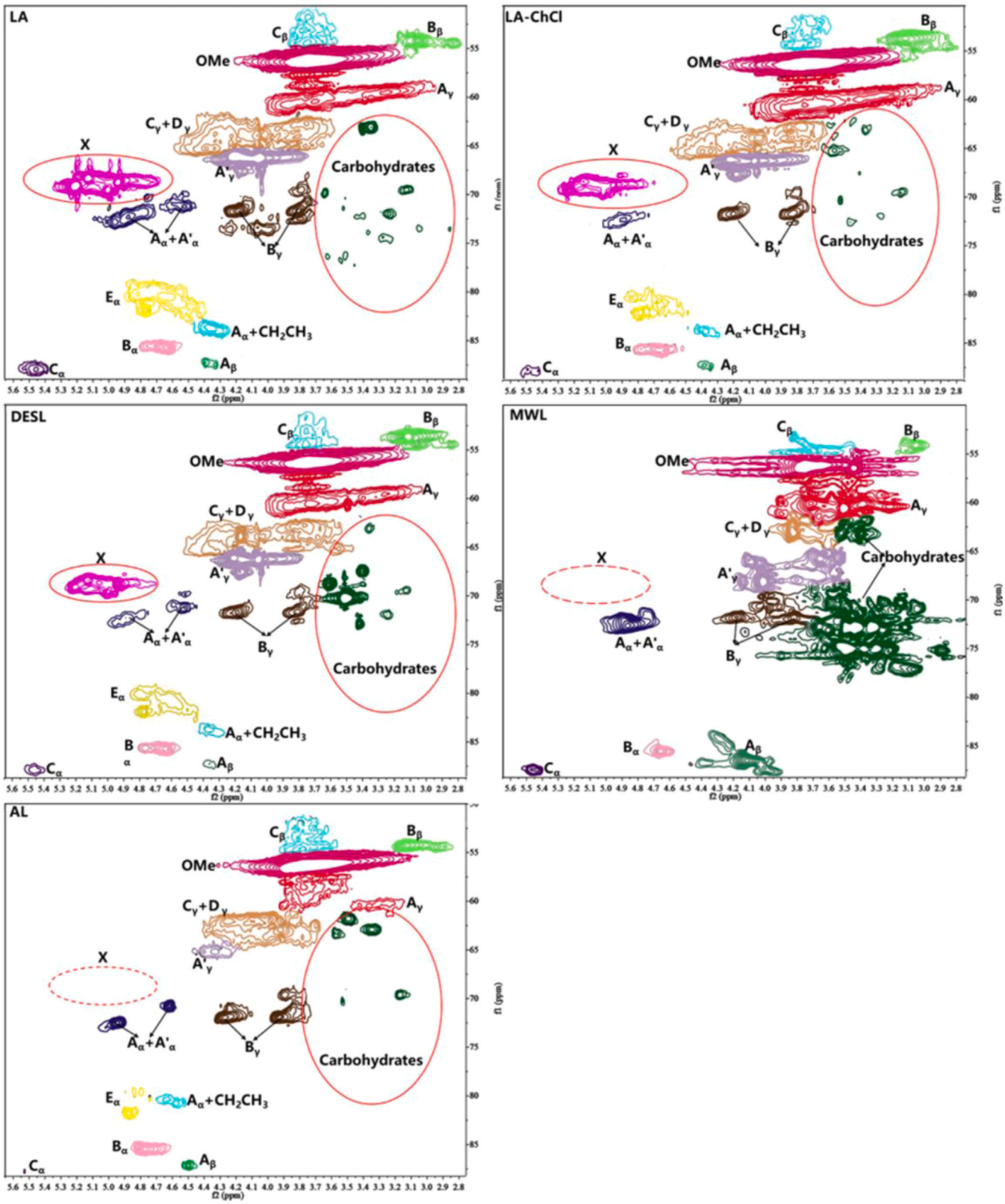


Fig. 9. HSQC spectra of lignin obtained in this work, side-chain regions (δ_C/δ_H 50–90/2.75–5.65). The conditions of LA and LA-ChCl: microwave heating at 140°C for 1 h. The conditions of LA-ChCl-PEG 400: microwave heating at 140°C for 30 min, the dosage of PEG 400 was 2.5 g. The same labels were applied to the ^{31}P analysis.

obviously after adding ChCl, which indicated that ChCl promoted the depolymerization of lignin. The Mw of lignin increased with the addition of PEG 400, which may be due to PEG 400 inhibiting lignin bond breakage by capturing hydrogen protons of LA. The Mw of lignin steadily declined as treatment time rose, and it initially fell and then climbed when the dose of the third component and chain length grew. These trends were negatively correlated with the delignification rate, which indicated that the reduction of Mw of lignin improved the

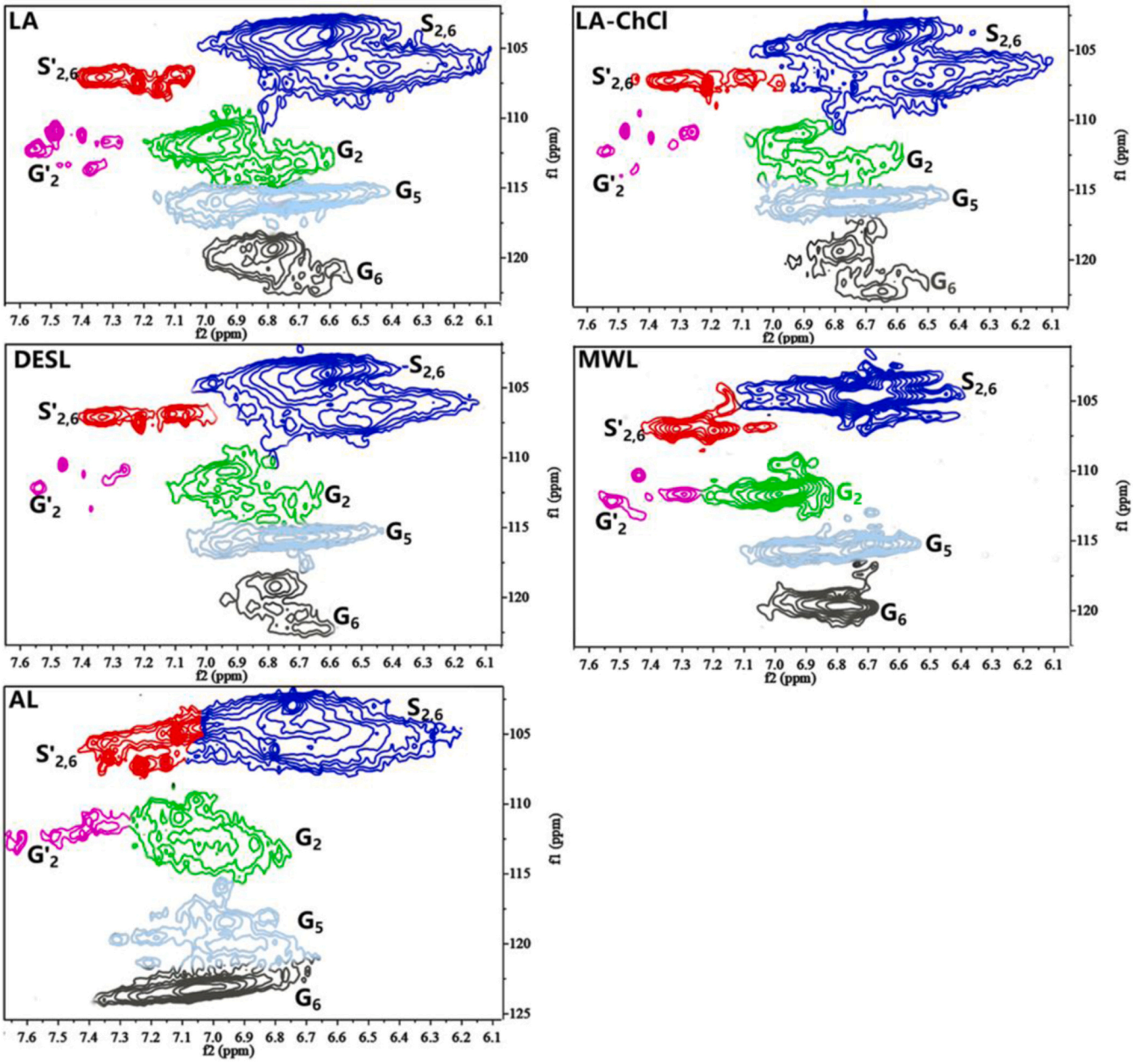


Fig. 10. HSQC spectra of lignin obtained in this work, aromatic regions (δ_C/δ_H 90–135/6.05–7.70).

delignification rate. Compared with MWL and AL, DES lignin has a smaller molecular weight and lower PDI, indicating that DESL has more excellent application value and better application prospects.

3.2.2. FTIR analysis

The FTIR spectra of lignin obtained in this work are shown in Fig. 8. The band assignments (Tab. S3) were determined based on previous studies (Casas et al., 2012). The result shows that the spectra of lignin samples obtained from DES treatment are almost identical.

Compared with MWL and AL, the DES lignin showed a strong signal at 1736 cm^{-1}, which belongs to the unconjugated carbonyl. During DES treatment, the β-O-4 bond may be cleaved, resulting in the production of unconjugated -ketone. Typical signs of the vibration of the primary aromatic chain emerged at 1606 cm^{-1}, 1511 cm^{-1}, and 1421 cm^{-1}. This meant that the aromatic skeleton was well retained in all of the lignin. The peak of 1328 cm^{-1}, 1124 cm^{-1}, and 832 cm^{-1} representing the syringyl (S) unit was evident in all lignin. In addition, the peaks of 1511 cm^{-1} and 1041 cm^{-1} are all from the guaiacyl (G) unit.

Table 4
The contents of the main functional groups in lignin used in this work.

Sample	S/G	β-O-4 (%)	β-β (%)	β-5 (%)
LA	2.234	25.104	10.818	4.573
LA-ChCl	2.330	11.680	8.607	2.820
LA-ChCl-PEG	2.332	17.133	9.589	2.580
MWL	2.524	72.740	13.881	2.421
AL	2.226	9.647	10.571	0.787

3.2.3. 2D-HSQC NMR analysis

The utilization of 2D-HSQC NMR represents a significant method in the examination of fundamental structural characteristics of lignin, including unit connectivity and unit types. The spectra of the examined lignin in the side-chain regions (C/H 50–90/2.75–5.65) and aromatic regions (C/H 90–135/6.05–7.70) are depicted in Figs. 9 and 10, respectively. Tab. S4 presents the comprehensive cross-signal distributions observed in the HSQC NMR spectra. Table 4 provides an overview

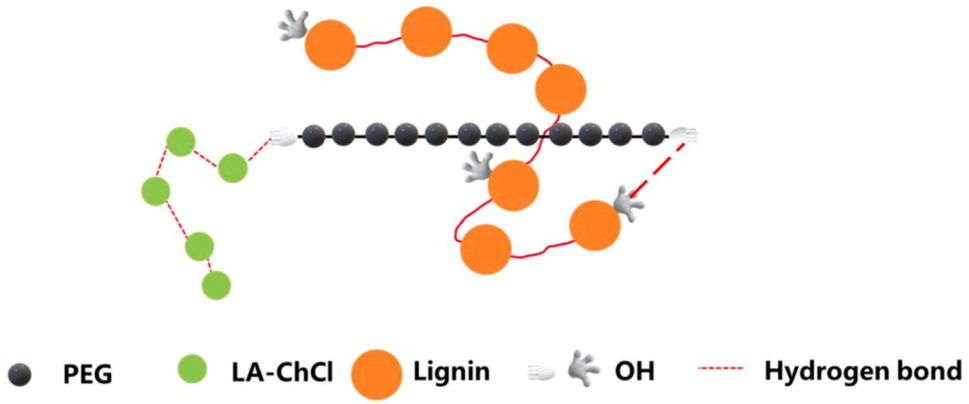


Fig. 11. The possible principle of PEG 400 increasing the solubility of lignin in solution.

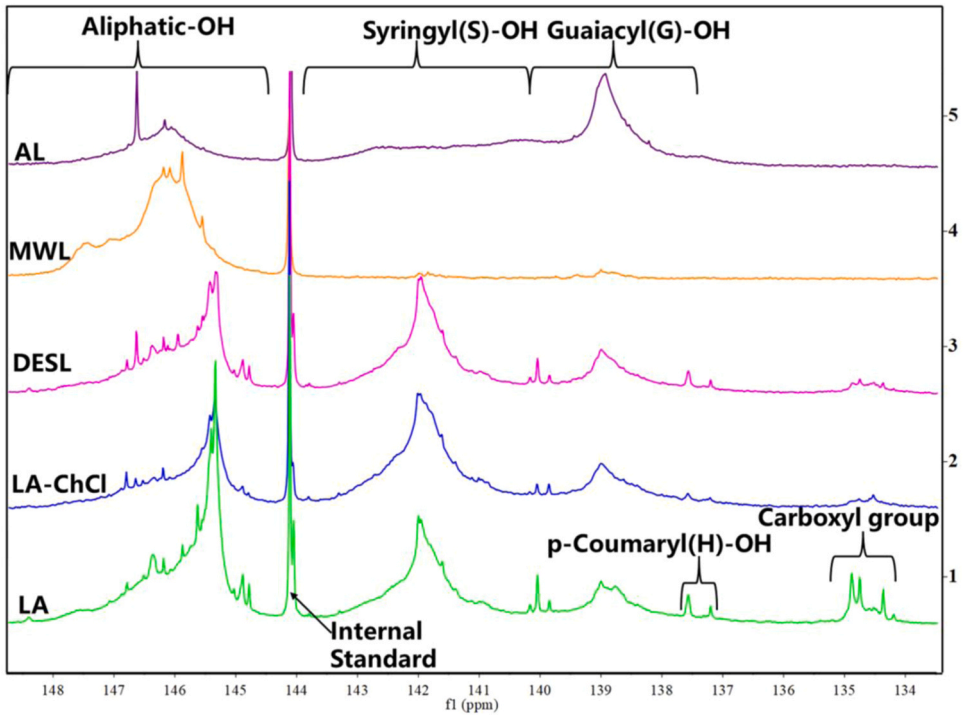


Fig. 12. ^{31}P NMR spectra of lignin obtained in this work.

of the constituent functional groups found in lignin. Additionally, Fig. S8 illustrates the substructures identified in the HSQC NMR spectra.

Compared with MWL, the content of the β-O-4 bond and the signal intensity of carbohydrates of all other lignins were decreased, which indicated that alkaline pulping and DES pretreatment mainly removed lignin by breaking the β-O-4 bond and lignin-carbohydrate complex (LCC) bond. In comparison to LA treatment, the utilization of ChCl demonstrated enhanced cleavage of three distinct types of lignin bonds, resulting in increased efficiency of lignin removal and a reduction in lignin molecular weight. These observations align with previous pretreatment investigations and the outcomes of molecular weight analysis. Compared with the LA-ChCl treatment for one hour, the lignin obtained by the LA-ChCl-PEG 400 treatment for 30 min had more bonding bonds and a stronger carbohydrate signal, but more lignin was removed by the LA-ChCl-PEG 400 treatment for 30 min. To find out the reason for this phenomenon, we tested the solubility of LA-ChCl and LA-ChCl-PEG 400 (10: 1 +5%) to AL according to our previously reported method (Liu et al., 2020), which was 22.71 wt% and 29.73 wt% at 60℃, respectively.

As shown in Fig. 11, the hydroxyl groups at both ends of long-chain PEG 400 can easily form hydrogen bonds with the hydroxyl groups of lignin. At the same time, PEG 400 can also form hydrogen bonds with LA-ChCl and has excellent mutual solubility in LA-ChCl. Therefore, PEG acts as a partial intermediate between lignin and LA-ChCl, which pulls lignin into the solution, thus increasing the solubility of the solution to lignin and improving the delignification rate. Compared with MWL, the S/G ratio of all lignin obtained after treatment decreased, which may be due to the removal of methoxy groups of some S-units. In addition, compared with MWL and AL, the lignin treated with LA-containing solvents showed a new signal X. This phenomenon has also appeared in our previous work (Liu et al., 2020), which may be due to the esterification reaction between lactic acid and the hydroxyl group of lignin (Morais et al., 2021; Lopes et al., 2020).

3.2.4. ^{31}P NMR analysis

The content of hydroxyl and carboxyl in lignin can be determined by quantitative ^{31}P NMR analysis. The spectrum of phosphorus and the contents of aliphatic hydroxyl, phenolic hydroxyl, and carboxyl of lignin

Table 5
The content of aliphatic hydroxyl, phenolic hydroxyl, and carboxyl of lignin used in this work.

Sample	Aliphatic hydroxyl (m mol/g)	S-OH (m mol/g)	G-OH (m mol/g)	H-OH (m mol/g)	Carboxyl (m mol/g)	Total phenolic hydroxyl (m mol/g)
LA	2.008	1.236	0.594	0.117	0.213	1.948
LA-ChCl	1.392	1.898	0.800	0.151	0.134	2.849
LA-ChCl-PEG 400	1.572	1.423	0.623	0.115	0.117	2.161
MWL	3.565	0.115	0.098	0.010	0.002	0.223

are shown in Fig. 12 and Table 5, respectively. Combined with 2D-HSQC NMR analysis, it was found that with the decrease of β-O-4 content (MWL > LA > LA-ChCl-PEG 400 > LA-ChCl), the aliphatic hydroxyl content gradually decreased and the phenolic hydroxyl content gradually increased. The reduction in aliphatic hydroxyl concentration can be ascribed to the acid-catalyzed

the dehydration process of hydroxyl groups on $C_{\alpha, \beta}$, or the entire side chain (Meng et al., 2018). The prominent cleavage of ether linkages in lignin may account for the observed increase in phenolic hydroxyl content. According to the research, the presence of phenolic hydroxyl groups is crucial for both the antioxidant and reactive properties of lignin (Xu et al., 2015). DES lignin has high phenolic hydroxyl content, which benefits its applications in antioxidants, chemical modification, and adhesives. The increase in the carboxyl group content of DES lignin may be due to the oxidation of hydroxyl groups. In the LA environment, the content of β-5 bond in lignin increased, indicating that lignin condensed, which was consistent with GPC analysis results. Moreover, adding ChCl and PEG-400 inhibited the condensation reaction of lignin.

3.2.5. *Analysis of simulated reactions by model compound*

The simulated reaction of the lignin model compound was carried out to better understand the reaction mechanism for lignin structural changes in the reaction system. The GC-MS analysis diagram of reaction products and the possible attribution of each peak in the diagram is shown in Fig. 13 and Tab. S5, respectively. The conversion rate of the lignin model compound was close to 100%, which indicates that the reaction system in this study has high efficiency for lignin conversion and has the potential for future transformation and application of DES lignin. The possible reaction route for the lignin model compound to produce main products (integral area > 1%) is given (Fig. 14). The integral area of PEG and LA was the biggest. Except for the signal of the solvent, guaiacol has the highest yield in monomer products, which supported the explanation for the increase of phenolic content in DESL. The appearance of G2 and G4 supports the description of the strong signal of DESL at 1736 cm^{-1} in the previous FTIR analysis. The appearance of G5 indicated that LA may react with the phenolic hydroxyl group of lignin. Combined with the emergence of new signal X in 2D-HSQC NMR analysis, it was speculated that signal X was G5.

In the whole reaction process, the aliphatic hydroxyl group in lignin and hydrogen proton in LA underwent a dehydration reaction, and the β-O-4 bond was cleaved to generate new monohydric phenol, which explained the reason why the aliphatic hydroxyl group decreased and the phenolic hydroxyl group increased in ^{31}P NMR analysis. A small amount of monohydric phenol containing N element was found in the reaction product (30.61 min), which indicated that choline ions did participate in the bond-breaking reaction of lignin, and further confirmed that choline ions promoted the removal of lignin. In addition, a small amount of ferulic acid (28.84 min) and lignin β-5 condensation product (30.70 min) was detected in the product, which provided a possible explanation for the increase of carboxyl content in DES lignin in the ^{31}P NMR analysis and the changes of β-5 content in lignin in 2D-HSQC NMR.

3.3. *The recovery and reuse of DES*

The recovery and reuse of DES were carried out three times. The recovery rate of DES was above 95% in each cycle. However, the delignification rate decreased rapidly, which were 89.73%, 68.54%, and 45.32% respectively. There are mainly two possible reasons. One reason is that in each cycle, the acidity of DES was reduced due to the bond breakage and dehydration reaction of lignin (Tab. S6). Another reason is that sugars produced by hydrolysis and part of lignin were dissolved in DES after each treatment, which significantly reduced the solubility of DES to lignin.

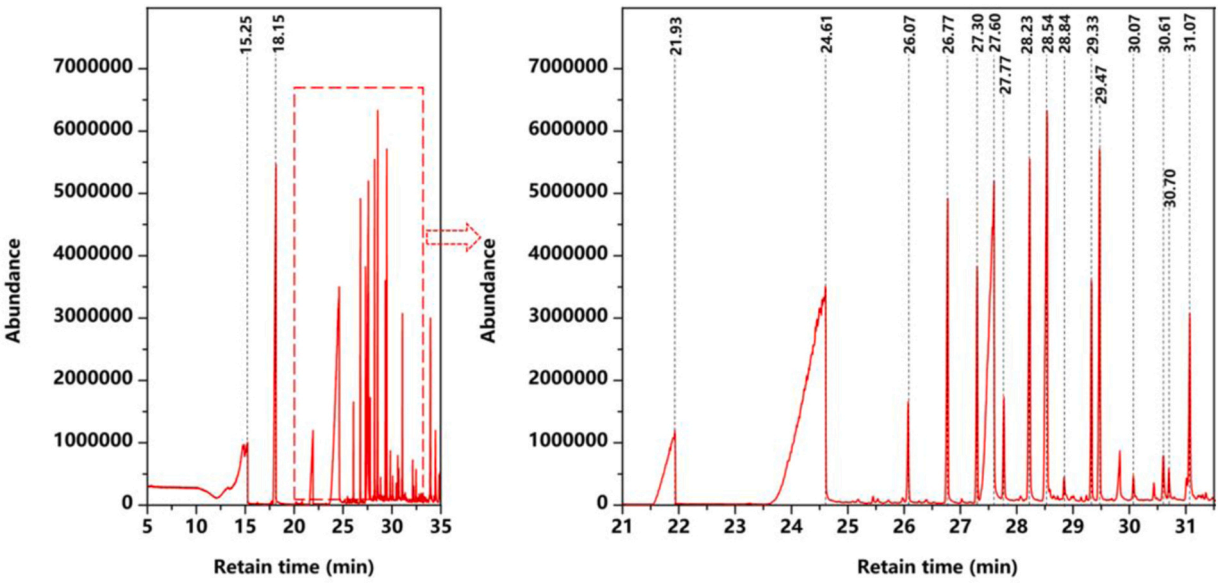


Fig. 13. GC-MS analysis of the products obtained from the simulated reaction. The conditions for stimulated reactions were: LA-ChCl-PEG 400 (10:1 +5%), microwave heating at 140℃ for 30 min, and the solid loading was 10 wt%.

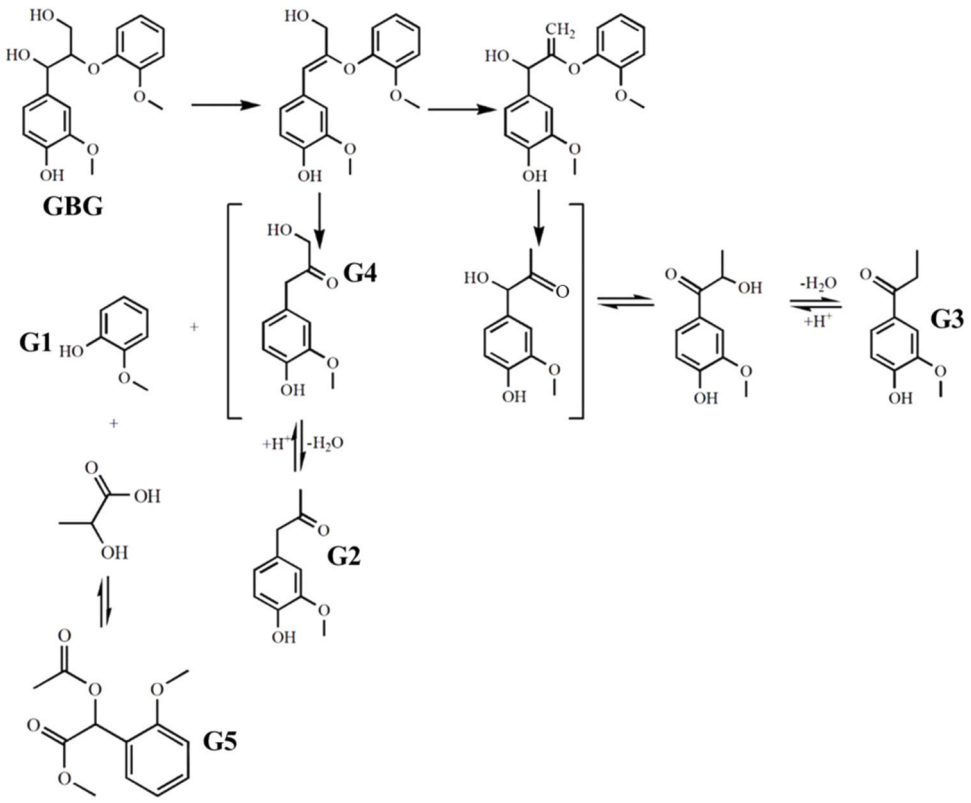


Fig. 14. Possible mechanism of the simulated reaction of lignin model compounds.

4. Conclusions

This work developed a strategy for rapid microwave-assisted pretreatment using a ternary deep eutectic solvent (LA:ChCl:PEG 400 (10:1 +5%)) for lignin fractionation of eucalyptus. At atmospheric pressure and 140℃, the delignification rate was 89.73% in 30 min. The hydroxyl group in LA increased the polarity and proton-donating ability of the solution. Carboxyl groups provided hydrogen protons to break lignin bonds. The increase in alkyl chain length reduced the acidity of DES. Choline ion promoted the cleavage of the lignin β-O-4 bond. PEG 400 protected cellulose from degradation. Moreover, PEG 400, as a solubilizing agent, improved the solubility of DES for lignin, promoted the dissolution of lignin, and thus improved the delignification rate. DESL had low molecular weight, a small polydispersity index, and a relatively rich content of β-O-4 and phenolic hydroxyl groups, which has great application value and good application prospects. Monophenols and Hibbert ketone were the main products of the simulated reaction of lignin model compounds.

CRediT authorship contribution statement

Yi Hou: Conceptualization, Funding acquisition, Project administration, Resources, Software, Supervision, Validation, Writing – review & editing. **Songqing Hu:** Conceptualization, Funding acquisition, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – review & editing. **Zhuang Liu:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.indcrop.2024.118297.

References

Liu, Y., Nie, Y., Lu, X., Zhang, X., He, H., Pan, F., Zhou, L., Liu, X., Ji, X., Zhang, S., 2019. Cascade utilization of lignocellulosic biomass to high-value products. Green. Chem. *21* (13), 3499–3535.

Rana, A.K., Scarpa, F., Thakur, V.K., 2022. Cellulose/polyaniline hybrid nanocomposites: design, fabrication, and emerging multidimensional applications. Ind. Crops Prod. *187*.

Rana, A.K., Guleria, S., Gupta, V.K., Thakur, V.K., 2023. Cellulosic pine needles-based biorefinery for a circular bioeconomy. Bioresour. Technol. *367*.

Rana, A.K., 2022. Green approaches in the valorization of plant wastes: recent insights and future directions. Curr. Opin. Green. Sustain. Chem. *38*.

Shen, X.-J., Wen, J.-L., Mei, Q.-Q., Chen, X., Sun, D., Yuan, T.-Q., Sun, R.-C., 2019a. Facile fractionation of lignocelluloses by biomass-derived deep eutectic solvent (DES) pretreatment for cellulose enzymatic hydrolysis and lignin valorization. Green. Chem. *21* (2), 275–283.

Haghighi Mood, S., Hossein Golfeshan, A., Tabatabaei, M., Salehi Jouzani, G., Najafi, G. H., Gholami, M., Ardjmand, M., 2013. Lignocellulosic biomass to bioethanol, a comprehensive review with a focus on pretreatment. Renew. Sustain. Energy Rev. (*27*), 77–93.

Ci, Y.-H., Yu, F., Zhou, C.-X., Mo, H.-E., Li, Z.-Y., Ma, Y.-Q., Zang, L.-H., 2020. New ternary deep eutectic solvents for effective wheat straw deconstruction into its high-value utilization under near-neutral conditions. Green. Chem. *22* (24), 8713–8720.

Tan, Y.T., Ngoh, G.C., Chua, A.S.M., 2018. Evaluation of fractionation and delignification efficiencies of deep eutectic solvents on oil palm empty fruit bunch. Ind. Crops Prod. *123*, 271–277.
Alvarez-Vasco, C., Ma, R., Quintero, M., Guo, M., Geleynse, S., Ramasamy, K.K., Wolcott, M., Zhang, X., 2016. Unique low-molecular-weight lignin with high purity extracted from wood by deep eutectic solvents (DES): a source of lignin for valorization. Green. Chem. *18* (19), 5133–5141.
da Costa Lopes, A.M., Gomes, J.R.B., Coutinho, J.A.P., Silvestre, A.J.D., 2020. Novel insights into biomass delignification with acidic deep eutectic solvents: a mechanistic study of beta-O-4 ether bond cleavage and the role of the halide counterion in the catalytic performance. Green. Chem. *22* (8), 2474–2487.
Smith, E.L., Abbott, A.P., Ryder, K.S., 2014. Deep Eutectic Solvents (DESs) and their applications. Chem. Rev. *114* (21), 11060–11082.
Schutyser, W., Renders, T., Van den Bosch, S., Koelewijn, S.F., Beckham, G.T., Sels, B.F., 2018. Chemicals from lignin: an interplay of lignocellulose fractionation, depolymerisation, and upgrading. Chem. Soc. Rev. *47* (3), 852–908.
Maugeri, Z., de Maria, P.D., 2012. Novel choline-chloride-based deep-eutectic-solvents with renewable hydrogen bond donors: levulinic acid and sugar-based polyols. RSC Adv. *2* (2), 421–425.
Zafarani-Moattar, M.T., Asadzadeh, B., Shekaari, H., 2016. Phase equilibrium of aqueous glycine plus choline chloride ionic liquid solutions. J. Solut. Chem. *45* (12), 1842–1856.
Zhou, M., Fakayode, O.A., Yagoub, A.E.A., Ji, Q., Zhou, C., 2022. Lignin fractionation from lignocellulosic biomass using deep eutectic solvents and its valorization. Renew. Sustain. Energy Rev. *156*.
Chen, Z., Ragauskas, A., Wan, C., 2020. Lignin extraction and upgrading using deep eutectic solvents. Ind. Crops Prod. *147*.
Tan, Y.T., Chua, A.S.M., Ngoh, G.C., 2020. Deep eutectic solvent for lignocellulosic biomass fractionation and the subsequent conversion to bio-based products - a review. Bioresour. Technol. *297*.
Wang, W., Lee, D.-J., 2021. Lignocellulosic biomass pretreatment by deep eutectic solvents on lignin extraction and saccharification enhancement: a review. Bioresour. Technol. *339*.
Hong, S., Li, H.-Y., Shen, X.-J., Sun, S.-N., Sun, Z., Yuan, T.-Q., 2022. Unveiling the migration and transformation mechanism of lignin in eucalyptus during deep eutectic solvent pretreatment. Chemsuschem *15* (15).
Shen, X.J., Wen, J.L., Mei, Q.Q., Chen, X., Sun, D., Yuan, T.Q., Sun, R.C., 2019b. Facile fractionation of lignocelluloses by biomass-derived deep eutectic solvent (DES) pretreatment for cellulose enzymatic hydrolysis and lignin valorization. Green. Chem. *21* (2), 275–283.
Chen, Y., Shen, K., He, Z., Wu, T., Huang, C., Liang, L., Fang, G., 2021. Deep eutectic solvent recycling to prepare high purity dissolving pulp. Cellulose *28* (18), 11503–11517.
Soares, B., Lopes, A.M. d C., Silvestre, A.J.D., Pinto, P.C.R., Freire, C.S.R., Coutinho, J.A. P., 2021. Wood delignification with aqueous solutions of deep eutectic solvents. Ind. Crops Prod. *160*.
Hong, S., Shen, X.-J., Pang, B., Xue, Z., Cao, X.-F., Wen, J.-L., Sun, Z.-H., Lam, S.S., Yuan, T.-Q., Sun, R.-C., 2020a. In-depth interpretation of the structural changes of lignin and formation of diketones during acidic deep eutectic solvent pretreatment. Green. Chem. *22* (6), 1851–1858.
Morais, E.S., Da Costa Lopes, A.M., Freire, M.G., Freire, C.S.R., Silvestre, A.J.D., 2021. Unveiling modifications of biomass polysaccharides during thermal treatment in cholinium chloride: lactic acid deep eutectic solvent. Chemsuschem *14* (2), 686–698.
Zhong, L., Wang, C., Yang, G., Chen, J., Xu, F., Yoo, C.G., Lyu, G., 2022. Rapid and efficient microwave-assisted guanidine hydrochloride deep eutectic solvent pretreatment for biological conversion of castor stalk. Bioresour. Technol. *343*.
Muley, P.D., Mobley, J.K., Tong, X., Novak, B., Stevens, J., Moldovan, D., Shi, J., Boldor, D., 2019. Rapid microwave-assisted biomass delignification and lignin depolymerization in deep eutectic solvents. Energy Convers. Manag. *196*, 1080–1088.
Chen, Z., Wan, C., 2018. Ultrafast fractionation of lignocellulosic biomass by microwave-assisted deep eutectic solvent pretreatment. Bioresour. Technol. *250*, 532–537.
Yan, Z., Wang, Z., Chen, Y., Liu, C., Liu, Y., Li, R., Si, M., Shi, Y., 2023. Preparation of lignin nanoparticles via ultra-fast microwave-assisted fractionation of lignocellulose using ternary deep eutectic solvents. Biotechnol. Bioeng. *120* (6), 1557–1568.
Chourasia, V.R., Bisht, M., Pant, K.K., Henry, R.J., 2022. Unveiling the potential of water as a co-solvent in microwave-assisted delignification of sugarcane bagasse using ternary deep eutectic solvents. Bioresour. Technol. *351*.
Ji, Q., Yu, X., Yagoub, A.E.-G.A., Chen, L., Zhou, C., 2020. Efficient removal of lignin from vegetable wastes by ultrasonic and microwave-assisted treatment with ternary deep eutectic solvent. Ind. Crops Prod. *149*.
Wen, J.-L., Sun, S.-L., Xue, B.-L., Sun, R.-C., 2013. Quantitative structural characterization of the lignins from the stem and pith of bamboo (Phyllostachys pubescens). Holzforschung *67* (6), 613–627.
Liu, Z., Hou, Y., Liu, C., Hu, S., 2023. Efficient and eco-friendly isolation and purification of lignin from black liquor with choline chloride-based deep eutectic solvents. Nord. Pulp Pap. Res. J.
Dwamena, A.K., Raynie, D.E., 2020. Solvatochromic parameters of deep eutectic solvents: effect of different carboxylic acids as hydrogen bond donor. J. Chem. Eng. Data *65* (2), 640–646.
Liu, Z., Hou, Y., Hu, S., Li, Y., 2020. Possible dissolution mechanism of alkali lignin in lactic acid-choline chloride under mild conditions. RSC Adv. *10* (67), 40649–40657.
Cui, J., Chen, R., Lei, L., Hou, Y., 2022. Green wood pulping processes with high pulp yield and lignin recovery yield by deep eutectic solvent and its aqueous solutions. Biomass Convers. Biorefinery.
Hong, S., Shen, X.-J., Xue, Z., Sun, Z., Yuan, T.-Q., 2020b. Structure-function relationships of deep eutectic solvents for lignin extraction and chemical transformation. Green. Chem. *22* (21), 7219–7232.
Kanbayashi, T., Miyafuji, H., 2016. Effect of ionic liquid treatment on the ultrastructural and topochemical features of compression wood in Japanese cedar (Cryptomeria japonica). Sci. Rep. *6* (1), 30147.
Yu, Y., Cheng, W., Li, Y., Wang, T., Xia, Q., Liu, Y., Yu, H., 2022. Tailored one-pot lignocellulose fractionation to maximize biorefinery toward versatile xylochemicals and nanomaterials. Green. Chem. *24* (8), 3257–3268.
Su, Y., Huang, C., Lai, C., Yong, Q., 2021. Green solvent pretreatment for enhanced production of sugars and antioxidative lignin from poplar. Bioresour. Technol. *321*.
Liu, Y., Chen, W., Xia, Q., Guo, B., Wang, Q., Liu, S., Liu, Y., Li, J., Yu, H., 2017. Efficient cleavage of lignin-carbohydrate complexes and ultrafast extraction of lignin oligomers from wood biomass by microwave-assisted treatment with deep eutectic solvent. Chemsuschem *10* (8), 1692–1700.
Wu, Y., Liu, Z., Ma, L., Guo, Z., Yang, Q., Wang, X., Hui, L., 2021. Structural characterization and evaluation of the antioxidant activity of DES-Lignin isolated from Cunninghamia lanceolata. Wood Sci. Technol. *55* (4), 1041–1055.
Li, T., Lyu, G., Liu, Y., Lou, R., Lucia, L.A., Yang, G., Chen, J., Saeed, H.A.M., 2017. Deep Eutectic Solvents (DESs) for the Isolation of Willow Lignin (Salix matsudana cv. Zhuliu). Int. J. Mol. Sci. *18* (11).
Casas, A., Virginia Alonso, M., Oliet, M., Rojo, E., Rodriguez, F., 2012. FTIR analysis of lignin regenerated from Pinus radiata and Eucalyptus globulus woods dissolved in imidazolium-based ionic liquids. J. Chem. Technol. Biotechnol. *87* (4), 472–480.
Lopes, A.M.D., Gomes, J.R.B., Coutinho, J.A.P., Silvestre, A.J.D., 2020. Novel insights into biomass delignification with acidic deep eutectic solvents: a mechanistic study of beta-O-4 ether bond cleavage and the role of the halide counterion in the catalytic performance. Green. Chem. *22* (8), 2474–2487.
Meng, X., Parikh, A., Seemala, B., Kumar, R., Pu, Y., Christopher, P., Wyman, C.E., Cai, C. M., Ragauskas, A.J., 2018. Chemical transformations of poplar lignin during cosolvent enhanced lignocellulosic fractionation process. ACS Sustain. Chem. Eng. *6* (7), 8711–8718.
Xu, J.-K., Sun, Y.-C., Sun, R.-C., 2015. Synergistic effects of ionic liquid plus alkaline pretreatments on eucalyptus: lignin structure and cellulose hydrolysis. Process Biochem. *50* (6), 955–965.